INCH-POUND
MIL-DTL-24787/3A(SH)
24 September 2020
SUPERSEDING
MIL-F-24787/3(SH)
3 September 1993

DETAIL SPECIFICATION SHEET

FITTINGS, END, TYPES C AND CL (O-RING SEAL UNION), REUSABLE, FOR FLEXIBLE HOSE ASSEMBLIES

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24787.

1. SCOPE

1.1 Scope. This specification covers reusable O-ring seal union end fittings (types C and CL) for hose sizes ranging from -4 to -32 for use with wire or synthetic fiber-reinforced rubber, flexible hose assemblies in accordance with MIL-DTL-24135 and MIL-DTL-24136.

1.2 Classification. The O-ring seal union end fittings are type C (straight nipple) and CL (90-degree elbow nipple) as identified on figure 1. For the corresponding hose type, see table I.

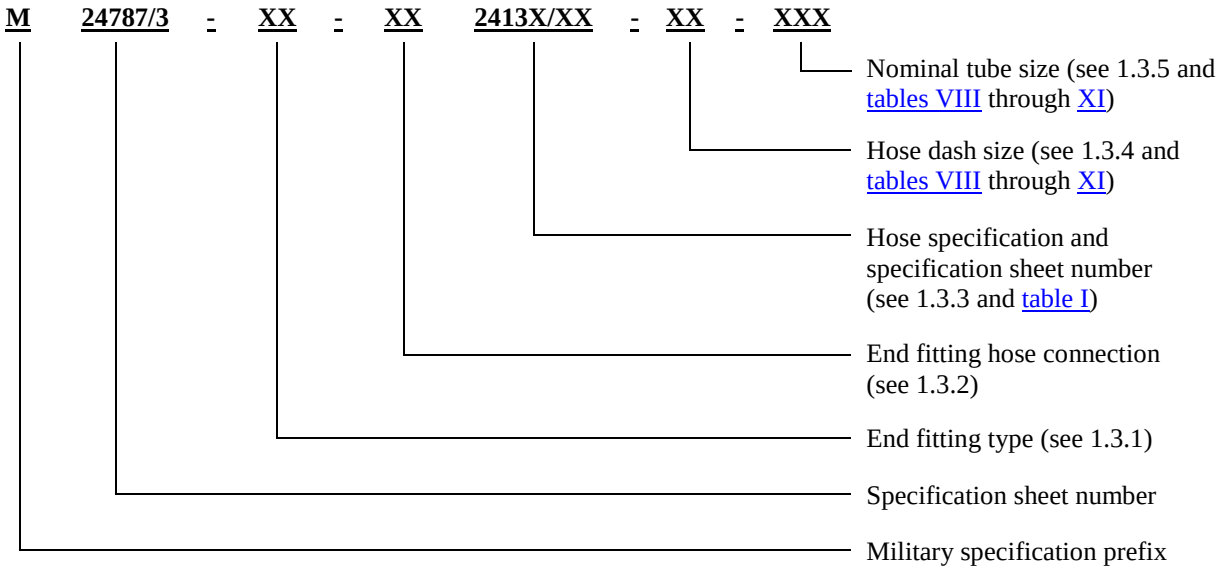
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AMSC N/A FSC 4730

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1.3 Part or identifying number (PIN). The PIN is a definitive part number that is formed by combining the applicable specification sheet number (M24787/3 for MIL-DTL-24787/3), a dash, end fitting type, a dash, end fitting hose connection, hose specification and applicable specification sheet number, a dash, hose dash size, a dash, and nominal tube size.



PIN example: M24787/3-CX-RB24135/01-32-150 describes a MIL-DTL-24787/3 end fitting, type C (straight nipple), with a ring and band hose connection, for use on MIL-DTL-24135/1, MIL-DTL-24135/5, and MIL-DTL-24135/13 hoses, with a hose size of -32, and a nominal tube size of 1½ inches.

1.3.1 End fitting type. End fitting types are identified in the PIN as follows:

End fitting type	Description	PIN cod
C	Straight nipple	CX
CL	90-degree elbow nipple	CL

1.3.2 End fitting hose connection. End fitting hose connections are identified in the PIN as follows:

Hose connection	PIN code
Solid Socket	SS
Ring and Band	RB
Bolt Together	BT

1.3.3 Hose specification and specification sheet number. Specification sheet numbers that only contain one digit will enter “0” as the first digit. For example, for specification sheet 1, the PIN code will be recorded as “01”. For end fittings categorized with multiple specification sheets, the first specification and applicable specification sheet will be used for the PIN. For example, for an end fitting that can be used with MIL-DTL-24135/1, MIL-DTL-24135/5, and MIL-DTL-24135/13, the PIN code will be recorded as “24135/01”.

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1.3.4 Hose dash size. Hose dash sizes in tables VIII through XI that only contain one digit will enter "X" as the second digit. For example, for a hose dash size of -4, the PIN code will be recorded as "4X". For hose dash sizes that contain two digits, all digits will be entered in the PIN. For example, for a hose dash size of -12, the PIN code will be recorded as "12".

1.3.5 Nominal tube size. When the tube size in tables VIII through XI is a value that does not contain a fraction, the inch size will be recorded in the first digit followed by two "X's". For example, for a nominal tube size of 2 inches, the PIN code will be recorded as "2XX". Tube sizes that only contain a dimension with a fraction (e.g., "½"), the PIN code will be entered as identified below. When the tube size is a value of 1 inch or above and contains a fraction, the "X's" below will be replaced with the inch value. For example, for a tube size of 1½ inches, the PIN code will be recorded as "150".

Tube size	PIN code
1/8	X13
¼	X25
3/8	X38
½	X50
¾	X75

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-T-1368 - Tube and Pipe, Nickel-Copper Alloy, Seamless and Welded

MIL-T-16420 - Tube, Copper-Nickel Alloy, Seamless and Welded, (Copper Alloy Numbers 715 and 706)

MIL-DTL-24135 - Hose, Synthetic Rubber, Wire Reinforced, for Flexible Hose Assemblies, General Specification for

MIL-DTL-24135/1 - Hose, Synthetic Rubber, Wire Reinforced (Double-Wire Braid or Double-Spiral and Single-Wire Braid), for Flexible Hose Assemblies (High Pressure, Sizes -4 through -32)

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MIL-DTL-24135/2	-	Hose, Synthetic Rubber, Wire Reinforced (Four-Ply Spiral-Wrapped or Double-Spiral and Single-Wire Braid), for Flexible Hose Assemblies (High Pressure, Sizes -6 through -16)
MIL-DTL-24135/5	-	Hose, Synthetic Rubber, Wire Reinforced (Double-Wire Braid), for Flexible Hose Assemblies (Perforated Cover, for High Pressure Gas, Sizes -4 through -32)
MIL-DTL-24135/9	-	Hose, Synthetic Rubber, Wire Reinforced (Four-Ply, Heavy Spiral-Wrapped), for Flexible Hose Assemblies (High Pressure, Sizes -8 through -32)
MIL-DTL-24135/10	-	Hose, Synthetic Rubber, Wire Reinforced (Single- or Double-Wire Braid), for Flexible Hose Assemblies (Medium Pressure, Sizes -4 through -48)
MIL-DTL-24135/12	-	Hose, Synthetic Rubber, High Performance, Wire Reinforced (Single-Wire Braid Over Single Synthetic Fiber Braid), for Flexible Hose Assemblies (Medium Pressure, High Temperature, Multi-Fluid, Sizes -4 through -40)
MIL-DTL-24135/13	-	Hose, Synthetic Rubber, High Performance, Wire Reinforced (Double-Wire Braid), for Flexible Hose Assemblies (High Pressure, High Temperature, Multi-Based Oil, Sizes -4 through -32)
MIL-DTL-24136	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies, General Specification for
MIL-DTL-24136/2	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies (High Pressure, Sizes -4 through -32)
MIL-DTL-24136/3	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies (Low Pressure, Sizes -4 through -32)
MIL-C-24723	-	Castings, Nickel-Copper Alloy
MIL-DTL-24787	-	Fittings, End, Reusable, for Flexible Hose Assemblies, General Specification for

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A494/A494M - Standard Specification for Castings, Nickel and Nickel Alloy

(Copies this document are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AMS6381 - Steel, Mechanical Tubing 0.95Cr – 0.20Mo (0.38 – 0.43C) (SAE 4140)

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. O-ring seal union end fittings shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

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3.2 O-ring seal union end fittings. O-ring seal union end fittings shall conform to the requirements of MIL-DTL-24787 and as specified herein.

3.2.1 Dimensions. End fitting dimensions shall be in accordance with figure 1 and tables VIII through XI (see 6.2). All dimensions in tables VIII through XI are recorded in inches.

3.2.2 Materials. End fitting materials shall be in accordance with figure 1 and tables VIII and XI. Nickel-based investment castings, if used, shall be in accordance with the material requirements of ASTM A494/A494M, with the exception of M-25S, and the welding and non-destructive testing (NDT) requirements of MIL-C-24723. M-25S made in accordance with ASTM A494/A494M shall be in accordance with MIL-C-24723 for mechanical property, welding, and NDT requirements.

3.2.3 End fitting availability. O-ring seal union end fittings (types C and CL) are available in the hoses and sizes indicated in table I.

TABLE I. O-ring seal union end fitting by hose specification sheet and size.

End fitting size (hose dash number)	Hose specification sheet				
	24135/1, /5, and /13	24135/2	24135/9	24135/10 (-4 to -32), /12, and 24136/3	24136/2
-6	X	X	--	X	X
-8	X	X	X	X	X
-10	--	--	--	X	--
-12	X	X	X	X	X
-16	X	X	X	X	X
-20	X	--	X	X	X
-24	X	--	X	X	X
-32	X	--	X	X	X

NOTE: X denotes where O-ring seal union end fittings are available by hose specification sheet and size.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-DTL-24787 and 4.4 of this specification.

4.3 Conformance inspection. Conformance inspection shall be in accordance with MIL-DTL-24787.

4.4 Sampling for qualification testing.

4.4.1 End fittings. Qualification testing of end fittings shall consist of end fitting samples as specified in table II. For qualification testing, end fittings shall be attached to the appropriate hose indicated in table II.

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4.4.2 Use on multiple hoses. Where end fittings are indicated for use on multiple associated hose specification sheets in table I, the basic qualification shall be to the hose associated detail specification having the highest pressure rating. At least one hose sample shall be selected from each additional associated specification sheet for the hydraulic impulse test.

TABLE II. End fitting samples required.

Hose size (dash number)	Number of end fitting samples required 1/	Appropriate hose specification sheet
-8	8	MIL-DTL-24135/1, /5, and /13
-32	10	
-6	8	MIL-DTL-24135/2
-16	10	
-16	8	MIL-DTL-24135/9
-32	10	
-8	8	MIL-DTL-24135/10 (-4 to -32), /12, and MIL-DTL-24136/3
-32	10	
-8	8	MIL-DTL-24136/2
-32	10	

NOTE:

1/ When qualifying fittings and hose, the number of end fittings for the small size increases to 10 and large size increases to 14 (see 4.4.4.2 or 4.4.4.3, MIL-DTL-24135 or MIL-DTL-24136, with associated hose detailed specification slant sheet).

4.4.3 Hose samples. Hose samples used for the qualification of O-ring seal union end fittings shall be qualified to the appropriate associated specification sheet and shall be not less than 15 inches long. The quantity of hose samples required shall be in accordance with table III. Where there is no qualified hose, the prospective end fitting supplier shall work together with a prospective hose supplier to qualify their respective products. (The hose supplier and end fitting supplier may be a single source.)

TABLE III. Hose samples required.

Hose size 1/ (table II)	Number of hose samples to be provided
Smallest	4
Largest	5

NOTE:

1/ When qualifying fittings and hose, the number of hoses for -8 increases to 5 and -32 increases to 7 (see 4.4.4.2 or 4.4.4.3 and MIL-DTL-24135 or MIL-DTL-24136, with the appropriate associated hose detailed specification slant sheet).

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4.4.4 Hose and end fitting mix. Where there are two or more qualified suppliers of a hose, or when the prospective end fitting supplier is also qualifying an end fitting, the end fitting and hose mix shall be in accordance with 4.4.4.1 to 4.4.4.3 herein, as appropriate. When qualifying a hose and end fitting, hose samples shall be coupled on one end with the end fitting being submitted for qualification and on the other end with a previously qualified end fitting.

4.4.4.1 Qualifying end fittings with multiple qualified hose suppliers. Where there are two or more qualified suppliers of a hose, the hose mix shall be in accordance with table IV.

TABLE IV. Hose mix when there are multiple qualified hose suppliers for end fitting qualification test.

Sample size [1/] (dash number)	Number of hoses to be provided [2/]	
	Qualified source #1	Qualified source #2
Smallest	2	2
Largest	3	2

NOTES:

1/ Satisfactory completion of qualification testing of the samples selected in accordance with tables II, III, and IV will qualify the manufacturer for all sizes in this associated specification sheet for the hose specification sheet(s). Qualification will be extended to all end fittings under the same hose specification sheet(s) under different associated end fitting specification sheets.

2/ See table III for appropriate hose sample.

4.4.4.2 Qualifying end fittings and hose with one qualified end fitting supplier and one qualified hose supplier. Where the prospective end fitting supplier is also qualifying a hose, and there is only one qualified supplier of end fittings, hose, or both, the end fitting and hose mix shall be in accordance with table V.

TABLE V. Hose and end fitting mix when there is one qualified supplier for combined qualification test.

Sample size [1/] (dash number)	Number of hose samples to be provided [2/]		Number of end fittings to be provided	
	Qualifier [3/]	Approved [4/] source	Qualifier [3/]	Approved [4/] source
Smallest	3	2	6	4
Largest	4	3	8	6

NOTES:

1/ Satisfactory completion of qualification testing of the samples selected in accordance with tables II, III, and V will qualify the manufacturer for all sizes in this associated specification sheet for the hose specification sheet(s). Qualification will be extended to all end fittings under the same hose specification sheet(s) under different associated end fitting specification sheets.

2/ See table III, note 1/, and MIL-DTL-24135 or MIL-DTL-24136, with the appropriate associated hose detailed specification slant sheet, for appropriate hose sample.

3/ Offeror of hose and end fittings or end fittings to be qualified.

4/ Approved source – previously qualified end fitting and hose, or end fitting or hose.

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4.4.4.3 Qualifying end fittings and hose with two or more qualified end fitting and qualified hose suppliers. Where the prospective end fitting supplier is also qualifying a hose and there are two or more qualified suppliers of end fittings, hose, or both, the end fitting and hose mix shall be in accordance with table VI.

TABLE VI. Hose and end fitting mix when there are two or more qualified suppliers for combined qualification test.

Sample size (dash number) 1/	Number of end fittings to be provided		
	Qualifier 2/	Approved 3/ source #1	Approved 4/ source #2
Smallest	6	2	2
Largest	8	3	3
Sample size (dash number) 1/	**Number of hoses to be provided**		
	Qualifier 2/	**Approved 3/ source #1**	**Approved 4/ source #2**
Smallest	3	1	1
Largest	4	2	1

NOTES:

1/ Satisfactory completion of qualification testing of the samples selected in accordance with tables II, III, and VI will qualify the manufacturer for all sizes in this associated specification sheet for the hose specification sheet(s). Qualification will be extended to all end fittings under the same hose specification sheet(s) under different associated end fitting specification sheets.

2/ See table III, note 1/, and MIL-DTL-24135 or MIL-DTL-24136, with the appropriate associated hose detailed specification slant sheet, for appropriate hose sample.

3/ Offeror of hose and end fittings or end fittings to be qualified.

4/ Approved source – previously qualified end fitting and hose, or end fitting or hose.

4.4.5 Test sequence. The sequence of tests on sample hose assemblies shall be as specified in table VII. If hoses are being qualified at the same time, the sequence of tests shall be in accordance with the applicable associated specification sheet for the hose being qualified. Whichever sequence is used, only those tests in table VII shall apply to O-ring seal end fittings.

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TABLE VII. Sequence of qualification tests on sample assemblies.

Test sequence (see MIL-DTL-24787)	Assembly sample numbers								
	Smallest size				Largest size				
	1	2	3	4	5	6	7	8	9
Proof pressure (see 4.5.1)	X	X	X	X	X	X	X	X	X
Oil circulation (see 4.5.2)	--	X	--	--	--	--	X	--	--
Impulse (see 4.5.3) 1/	--	--	--	X	X	--	X	--	--
Burst pressure (see 4.5.4)	X	--	X	--	--	X	--	X	-

NOTE:

1/ Where the end fitting group being qualified is used on multiple hoses (see table II), qualification will be extended to the use of the end fitting on additional hoses provided that at least one sample hose assembly from each identified hose specification sheet is made up and submitted to the hydraulic impulse test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. O-ring seal union end fittings to this specification are intended for use in shipboard or shore-to-ship piping systems where pressures do not exceed the flange working pressure and temperatures do not exceed that of the applicable hose.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification sheet.

b. Quantity of end fittings.

c. Hose dash number and nominal tube size (see 3.2.1).

d. End fitting type, class, hose specifications, and size (hose by pipe or tube) (see 3.2.3).

e. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24787 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

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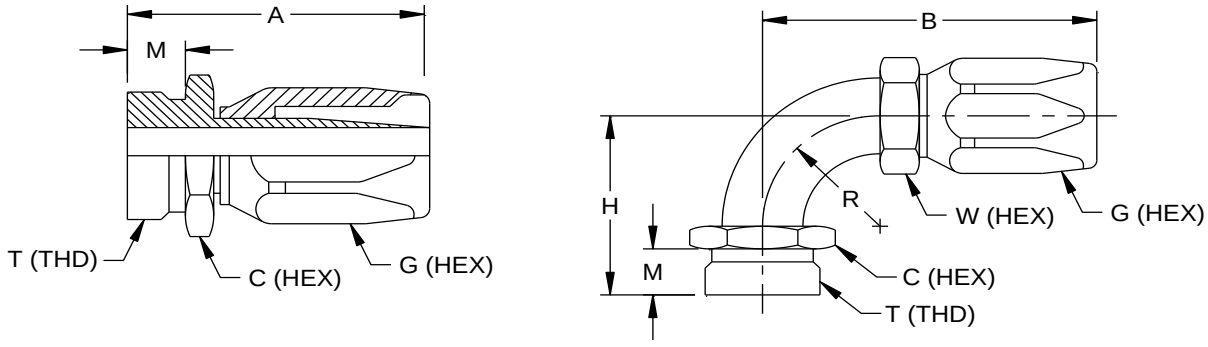
6.4 Subject term (key word) listing.

Synthetic fiber

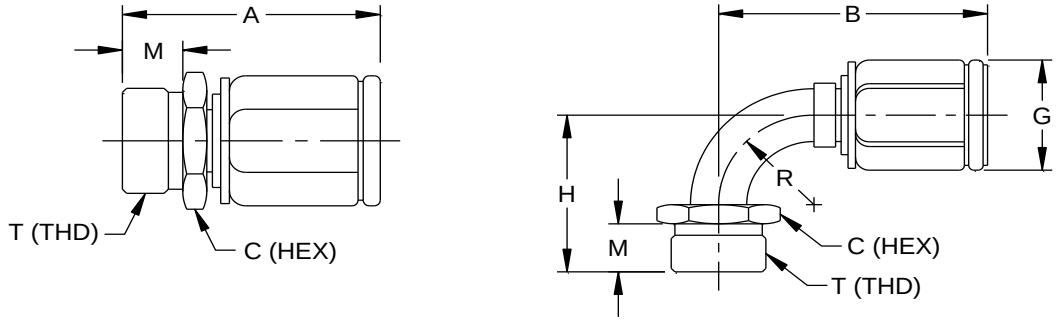
Wire

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

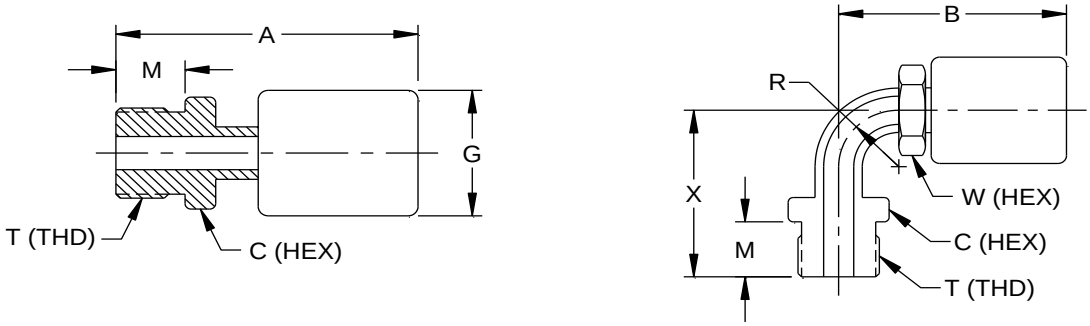
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TYPE C AND CL - - - - - SOLID SOCKET



TYPE C AND CL - - - - - RING AND BAND



TYPE C AND CL - - - - - BOLT TOGETHER

FIGURE 1. Type C and CL end fittings.

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TABLE VIII. Dimensions for MIL-DTL-24135/1, /2, /5, and /13 solid socket types C and CL.

Hose size (dash number)	Tube size (nom)	T Thread CL-2B	A max	C nom	G max	M ref	B max	H max	R min	W nom
-6	¼	1 - 3/16 - 12 UN	3.25	1.25	1.25	0.62	3.35	2.25	0.50	0.81
-6	⅜	1 - ⅜ – 12 UNF	3.25	1.44	1.25	0.62	X	X	X	X
-8	⅜	1 - ⅜ – 12 UNF	3.50	1.44	1.50	0.62	3.81	2.50	0.75	0.81
-12	½	1-¾ - 12 UN	4.00	1.81	1.75	0.62	4.75	3.25	1.25	1.20
-16	¾	2-12 UN	4.50	2.12	2.25	0.62	5.75	3.75	1.50	1.44
-16	1	2 - 5/16 - 12 UNS	4.75	2.38	2.25	0.75	X	X	X	X
-16	1¼	2-¾ - 12 UN	5.00	2.88	2.25	0.75	X	X	X	X
-20	1	2 - 5/16 - 12 UNS	5.25	2.38	2.75	0.75	6.75	4.25	1.75	1.62
-20	1¼	2-¾ - 12 UN	5.50	2.88	2.75	0.77	X	X	X	X
-24	1¼	2-¾ - 12 UN	5.25	2.88	3.00	0.75	7.00	5.00	2.00	1.88
-32	1½	3 - 1/16 – 12 UN	6.00	3.12	3.50	0.88	8.00	5.75	2.50	2.50
-32	2	3-¾ - 12 UN	6.00	3.88	3.50	0.88	8.25	6.00	2.50	2.50

NOTES:

1. End fitting assembly materials for socket and hose nipple shall be in accordance with QQ-N-281 or QQ-N-286.
2. Socket (body) alternate material shall be in accordance with MIL-C-24723 or ASTM A494/A494M. If ASTM A494/A494M is used, see 3.2.2 for additional guidance.
3. The 90-degree elbow for sizes -6 through -12 shall be in accordance with MIL-T-1368 and for sizes -16 through -32, to MIL-T-16420.
4. The socket shall be a one-piece design.

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TABLE IX. Dimensions for MIL-DTL-24135/10 (-4 to -32), /12, and MIL-DTL-24136/3 solid socket types C and CL.

Hose size (dash number)	Tube size (nom)	T Thread CL-2B	A max	C nom	G max	M ref	B max	H max	R ref	W nom
-6	⅛	1–14 UNF	2.06	1.06	0.81	0.50	2.45	1.30	0.50	0.56
-8	¼	1 - 3/16 - 12 UN	2.54	1.25	0.94	0.62	3.05	1.68	0.75	0.75
-10	⅜	1 - ⅜ - 12 UNF	2.75	1.44	1.12	0.62	3.57	1.99	1.00	0.81
-12	½	1 - ⅜ - 12 UNF	3.12	1.81	1.25	0.61	4.15	2.31	1.25	1.00
-16	¾	2 - 12 UN	2.80	2.12	1.44	0.62	3.93	2.68	1.50	1.38
-20	1	2 - 5/16 - 12 UNS	3.14	2.38	1.75	0.75	4.27	3.17	1.75	1.62
-24	1¼	2 -¾ - 12 UN	3.23	2.88	1.94	0.75	4.65	3.42	2.00	1.81
-32	1½	3 - 1/16 – 12 UN	3.89	3.12	2.44	0.88	5.68	4.05	2.50	2.38

NOTES:

1. End fitting assembly materials for socket and hose nipple shall be in accordance with QQ-N-281 or QQ-N-286.
2. Socket (body) alternate material shall be in accordance with MIL-C-24723 or ASTM A494/A494M. If ASTM A494/A494M is used, see 3.2.2 for additional guidance.
3. The 90-degree elbow for sizes -6 through -12 shall be in accordance with MIL-T-1368 and for sizes -16 through -32, to MIL-T-16420.
4. The socket shall be a one-piece design.

TABLE X. Dimensions for MIL-DTL-24135/9, ring and band socket, types C and CL.

Hose size (dash number)	Tube size (nom)	T Thread CL-2B	A max	C nom	G max	M ref	B max	H max	R ref
-8	⅜	1 - ⅜ - 12 UNF	3.63	1.43	1.60	0.62	4.45	2.05	0.75
-12	½	1 - ¾ - 12 UN	4.60	1.81	1.78	0.62	5.90	2.50	1.25
-16	1	2 - 5/16 - 12 UNS	5.17	2.38	2.16	0.75	6.40	3.10	1.50
-20	1	2 - 5/16 - 12 UNS	5.10	2.38	2.88	0.75	X	X	X
-20	1¼	2 - ¾ - 12 UN	5.10	2.88	2.88	0.75	6.44	3.32	1.75
-24	1¼	2 - ¾ - 12 UN	5.42	2.88	3.16	0.75	X	X	X
-24	1½	3 - 1/16 - 12 UN	5.68	3.12	3.16	0.88	7.25	4.94	2.00
-32	2	3 - ¾ - 12 UN	6.01	3.88	3.75	0.88	8.95	4.94	2.50

NOTES:

1. End fitting assembly materials for the hose nipple shall be in accordance with QQ-N-281 or QQ-N-286.
2. Sockets for sizes -8, -12, and -16 shall be one-piece design. Sockets for sizes -20 and larger may be a one-piece or multiple segment design, not exceeding the envelope dimensions.
3. Material for the sockets shall be in accordance with QQ-N-281, QQ-N-286, MIL-DTL-24723, or ASTM A494/A494M for sizes -20 and larger, AISI/SAE 4140 steel shall be in accordance with SAE AMS6381, as applicable. If ASTM A494/A494M is used, see 3.2.2 for additional guidance.
4. The 90-degree elbow tubing for sizes -8 and -12 shall be in accordance with MIL-T-1368 and for sizes -16 and larger, to MIL-T-16420.

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TABLE XI. Dimensions for MIL-DTL-24136/2, bolt together socket, types C and CL.

Hose size (dash number)	Tube size (nom)	T Thread CL-2B	A max	C nom	G max	M ref	X max	B max	H max	R ref	W nom
-6	¼	1 - 3/16 – 12 UN	3.25	1.25	1.25	0.62	2.47	3.25	2.25	0.50	0.94
-6	3/8	1 - 3/8 - 12 UNF	3.25	1.44	1.25	0.62	2.47	X	X	X	X
-8	3/8	1 - 3/8 - 12 UNF	3.50	1.44	1.50	0.62	3.00	3.75	2.50	0.75	1.00
-12	3/8	1 - 3/8 - 12 UNF	3.75	1.44	1.75	0.62	3.50	X	X	X	X
-12	½	1 - ¾ - 12 UN	4.00	1.81	1.75	0.62	3.50	4.75	3.25	1.25	1.31
-16	½	1 - ¾ - 12 UN	4.25	1.81	2.25	0.62	4.81	X	X	X	X
-16	¾	2- 12 UN	4.50	2.12	2.25	0.62	4.81	5.75	3.75	1.50	1.62
-16	1	2 - 5/16 - 12 UNS	4.75	2.38	2.25	0.75	4.81	X	X	X	X
-16	1¼	2 - ¾ - 12 UN	5.00	2.88	2.25	0.75	4.81	X	X	X	X
-20	1	2 - 5/16 - 12 UNS	5.25	2.38	2.75	0.75	4.81	6.75	4.25	1.75	2.00
-20	1¼	2 - ¾ - 12 UN	5.50	2.88	2.75	0.75	4.50	X	X	X	X
-24	1¼	2 - ¾ - 12 UN	5.25	2.88	3.00	0.75	5.38	7.00	5.00	2.00	2.00
-32	1½	3 - 1/16 - 12 UN	6.00	3.12	3.50	0.88	5.81	8.00	5.75	2.50	2.75
-32	2	3 - ¾ - 12 UN	6.00	3.88	3.50	0.88	5.81	8.25	6.00	2.50	2.75

NOTE: End fitting assembly materials shall be in accordance with QQ-N-281 or QQ-N-286.

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CONCLUDING MATERIAL

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